**FieldComms** is a self-contained emergency communications server built on a Raspberry Pi 5 for McHenry County RACES, ARES, and Starcom operations. Any smartphone, tablet, or laptop connects to the **EMCOMM-NET** Wi-Fi access point and reaches the full dashboard at **http://192.168.50.1** — no internet, no app installation, no per-device configuration required. All 30 tools, all ICS forms, all reference materials, and all offline maps remain fully functional when the site has no internet connectivity.

| ■ AMATEUR RADIO | ■ STARCOM / PUBLIC SAFETY | ■ ICS INCIDENT COMMAND |
|---|---|---|
| • Net Control Logger — FCC auto-fill | • Starcom Net Logger — Radio ID / unit-based | • Command — objectives, safety, staff |
| • Callsign Lookup — 800K licensees offline | • Weather Net — storm spotter check-in | • Operations — T-card board, assignments |
| • Observer Mode — read-only agency view | • SAR Net — search and rescue radio ops | • Planning — IAP tracker, resource status |
| • APRS Tactical Map — Graywolf + YAAC | • Resource Tracking Map — unit placement | • Logistics — ICS-205 comms plan, supplies |
| • HF Propagation — band conditions live | • Resource Board — 5-state status cycle | • Finance/Admin — cost, time, procurement |
| • Repeater Database — ARES/band filters | • Member Roster — certifications, activations | • Planning P — 15-phase planning cycle guide |
| • Dead Man's Switch — net inactivity alert | • Facilities Directory — EOC, shelters, hospitals | • ICS-213 / ICS-214 / ICS-309 forms |
| • NTS Radiogram — ARRL formatted traffic | • Multiple simultaneous nets on one screen | • Winlink Form Import — XML to incident |
| • JS8Call — HF keyboard digital messaging | • Observer link for served agencies | • Incident types: Disaster, HazMat, SAR, MCI |
| • Pat Winlink — backup email over radio | • ICS platform integration from Starcom mode | • Multi-user — all sections see live data |

## ■ Connectivity — Dual WAN with Automatic Failover

| InstyConnect Cellular (Primary WAN) | Starlink Satellite (Automatic Failover) | AMPRNet / 44Net (Amateur Radio IP) |
|---|---|---|
| InstyConnect Drum omnidirectional antenna + Switchblade directional backup. T-Mobile + Verizon dual-carrier. Pauses at $5/month between activations. | Automatic failover when cellular drops. ASUS RT-BE58 Go manages failover with no operator action required. All FieldComms features remain active during satellite operation. | Dedicated Raspberry Pi 5 gateway. WireGuard tunnel to amprgw.ampr.org. Routes 44.0.0.0/8 for all EMCOMM-NET devices. AMPRNet allocation required from portal.ampr.org. |

## ■ Hardware Platform

| COMPONENT | SPECIFICATION | ROLE |
|---|---|---|
| FieldComms Server | Raspberry Pi 5 16 GB · Pironman MAX 5 · 2× 1 TB NVMe RAID 1 | Runs all 30 web pages, 12 Python services, FCC database, offline maps, Kiwix library |
| 44Net Gateway | Raspberry Pi 5 16 GB · Argon NEO 5 · 256 GB M.2 SSD · Pi OS Desktop | Dedicated AMPRNet WireGuard gateway · routes 44.0.0.0/8 for all devices |
| Wi-Fi Access Point | ASUS RT-BE58 Go Wi-Fi 7 · Dual WAN (cellular primary + satellite failover) | EMCOMM-NET SSID · DHCP server · WAN gateway with automatic failover |
| Network Switch | UniFi Switch Lite 16 PoE · 16-port GbE · 8× PoE · 2× SFP uplink | Wired distribution · both Pis, workstations, printer, spare ports |
| Primary Cellular WAN | InstyConnect Drum (omni) + Switchblade (directional backup) · 5G/LTE | T-Mobile + Verizon dual-carrier · PoE Ethernet to ASUS WAN port |
| Satellite WAN Backup | Starlink Standard or Flat HP dish · auto-failover via ASUS USB WAN | Secondary internet when cellular unavailable · CGNAT · no inbound connections |
| Radio Station | Icom IC-7300 · Windows laptop · Winlink Express + VARA HF · JS8Call | HF digital voice and data · Winlink email over radio · IC-7300 via single USB |
| Operator Workstations | Raspberry Pi 500 / 500+ ×4 · Raspberry Pi Monitor 15.6" ×4 | Browser-based · one per operator station · USB-C powered from monitor |

### Reference & Administration

• Kiwix Offline L

### How to Access

501(c)(3) Grant Eligible · Funding Sources: FEMA BRIC · ARPA-E · ARRL Foundation · ARES Special Fund · DHS BRIC Nonprofit Security Grant Program · Illinois Homeland Security